

Streptobacillary Rat-Bite Fever (MONDO:0020533): A Comprehensive Disease Characterization

Summary

Streptobacillary rat-bite fever (RBF) is a rare, acquired zoonotic bacterial infection caused by the fastidious Gram-negative bacillus *Streptobacillus moniliformis* (with the recently split *Streptobacillus notomys* recognized as an additional human pathogen). It is transmitted by rodent bites and scratches, by handling rodents or their excreta, or by ingesting rodent-contaminated food or water — the ingestion form is historically termed **Haverhill fever** (erythema arthriticum epidemicum). After a typical incubation of 3–10 days, the disease produces an acute systemic illness with the classic triad of **fever, an acral (hands/feet) rash, and migratory polyarthritis**. Because it is an infectious rather than a genetic disease, all host-genetic, inheritance, penetrance, and gene-therapy dimensions of the research template are **Not Applicable**.

The central clinical lesson from this investigation is that **outcome hinges almost entirely on time-to-diagnosis, not on host factors**. RBF is exquisitely susceptible to penicillin and is reliably curable when recognized early, yet its nonspecific initial presentation and a difficult-to-culture organism together produce a real risk of diagnostic delay. Untreated or late-treated disease can progress to septic arthritis, osteomyelitis, infective endocarditis, and fatal multi-organ sepsis, with reported untreated case-fatality of **10–13%**. Diagnosis rests on exposure history plus recovery/detection of the organism, aided increasingly by molecular methods (16S rRNA PCR, triplex/multiplex qPCR, metagenomic next-generation sequencing) that circumvent the organism's fastidious growth and inhibition by the anticoagulant sodium polyanethole sulfonate (SPS) in blood-culture bottles.

This report consolidates 15 confirmed findings across 28 reviewed papers. It covers the disease's identifiers and synonyms, etiology and transmission, phenotypes, the (non-genetic) molecular/host picture, environmental and occupational determinants, a stepwise pathophysiological causal chain, anatomical involvement, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, and its comparative/veterinary biology.

Prevention is behavioral and occupational — no human vaccine exists — and centers on avoiding rodent exposure, wound hygiene, and consideration of post-exposure antibiotic prophylaxis.

1. Disease Information

Overview. Streptobacillary rat-bite fever is a systemic febrile zoonosis caused by *Streptobacillus moniliformis*, "a systemic illness classically characterized by fever, rigors, and polyarthralgias" (PMID: 17223620). It is one of two classic forms of rat-bite fever; the other, more common in Asia, is caused by *Spirillum minus* (spirillary RBF / sodoku). This report concerns the streptobacillary form.

Key identifiers.

Resource	Identifier
Mondo	MONDO:0020533
ICD-10	A25.1 (Streptobacillosis)
ICD-11	1B93.0
MeSH	D011906 (Rat-Bite Fever)
Disease Ontology	DOID:0050144
Causative organism (NCBI Taxonomy)	txid34105 (<i>S. moniliformis</i>)
OMIM / Orphanet genetic entry	None (not a Mendelian disease)

Synonyms and alternative names. Streptobacillary fever; streptobacillosis; **Haverhill fever** and **erythema arthriticum epidemicum** (the food/water-borne, ingestion-acquired form); epidemic arthritic erythema. "Rat-bite fever" is the umbrella clinical term shared with the spirillary form.

Source of information. The knowledge base entry is derived from **aggregated disease-level resources** — case reports, case series, and narrative reviews — rather than large structured EHR cohorts. The disease is rare and, in most jurisdictions, **not nationally reportable**, so "it may be more prevalent than currently recognized" (PMID: 42623631).

2. Etiology

Primary cause — infectious. The disease is caused by infection with *Streptobacillus moniliformis*, a fastidious, pleomorphic, Gram-negative bacillus that is a **commensal of the rat naso-/oropharynx**. A second species, *S. notomytis*, is now recognized as "a distinct species from *Streptobacillus moniliformis* — the primary causative agent of rat-bite fever" and "has been implicated in rare human infections" (PMID: 40472936).

Risk factors (environmental / behavioral). - **Rodent exposure** — bites, scratches, handling, or contact with rodent excreta. "Rat-bite fever follows exposure to contaminated bodily fluids of infected rodents" (PMID: 34323361). - **Occupational exposure** — pet-shop employees, laboratory-animal workers, and others in contact with rodents. A fatal case occurred in a 24-year-old pet-shop employee who "contracted the disease through a minor superficial finger wound on a contaminated rat cage" (PMID: 15029568). - **Age and social factors** — young children and people of low socioeconomic status are overrepresented; RBF has "potentially lethal course in a vulnerable population (children and low socioeconomic class)" (PMID: 34672901). - **Pet-rodent ownership** — an increasing driver as pet rats become more common.

Genetic risk / protective factors. Not applicable. No causal genes, susceptibility loci, modifier genes, protective alleles, or gene–environment interactions are described for this acquired infection.

3. Phenotypes

The hallmark is the **clinical triad**: physicians should consider RBF "when fever, rash, and exposure to rats are part of the patient's history" (PMID: 11724672); it presents as a "clinical triad of symptoms, fever, rash and arthritis" (PMID: 39267964).

Phenotype	Type	Characteristics	Suggested HPO term
Fever / rigors	Symptom (constitutional)	Abrupt onset, often high; near-universal	HP:0001945 (Fever)
Rash (maculopapular, petechial, pustular, purpuric; acral)	Physical manifestation	Characteristically on hands and feet, including palms/soles; hemorrhagic pustules/purpuric lesions	HP:0000988 (Skin rash)
Migratory polyarthritis / polyarthralgia	Clinical sign	Asymmetric, migratory; can progress to frank septic arthritis	HP:0005764 (Migratory arthritis) / HP:0002829 (Arthralgia)
Myalgia	Symptom	Common in acute phase	HP:0003326 (Myalgia)
Headache	Symptom	Frequent	HP:0002315 (Headache)
Leukocytosis / neutrophilia	Laboratory abnormality	Variable; WBC may be normal	HP:0001974 (Leukocytosis)
Thrombocytopenia	Laboratory abnormality	Reported in severe/ atypical cases	HP:0001873 (Thrombocytopenia)
Elevated CRP	Laboratory abnormality	Nonspecific inflammatory marker	HP:0011227 (Elevated C- reactive protein)

RBF presents with "unspecific symptoms, including fever, arthralgia, and polymorphous skin lesions" ([PMID: 34323361](#)). **Onset** is adult or pediatric (any age), **severity** ranges from mild self-limited illness to fatal sepsis (variable), and **progression** is acute and may be episodic/relapsing if untreated. **Quality-of-life impact** during acute illness is substantial (fever, disabling polyarthritis) but, with prompt treatment, is typically fully reversible; formal EQ-5D/SF-36 data are not available for this rare disease.

4. Genetic / Molecular Information

Not applicable. Streptobacillary RBF is an acquired bacterial infection with **no human causal genes, no inheritance pattern, no pathogenic germline or somatic variants, no disease-defining epigenetic mechanism, and no chromosomal abnormality**. Sections on causal genes, pathogenic variants (classification, allele frequency, somatic vs germline), modifier

genes, and host epigenetics are Not Applicable. The relevant "molecular" entity is the pathogen genome itself (NCBI:txid34105), not a host locus. The clinical characterization emphasizes an acute systemic **infectious** course rather than a genetic disease (PMID: 17223620).

5. Environmental Information

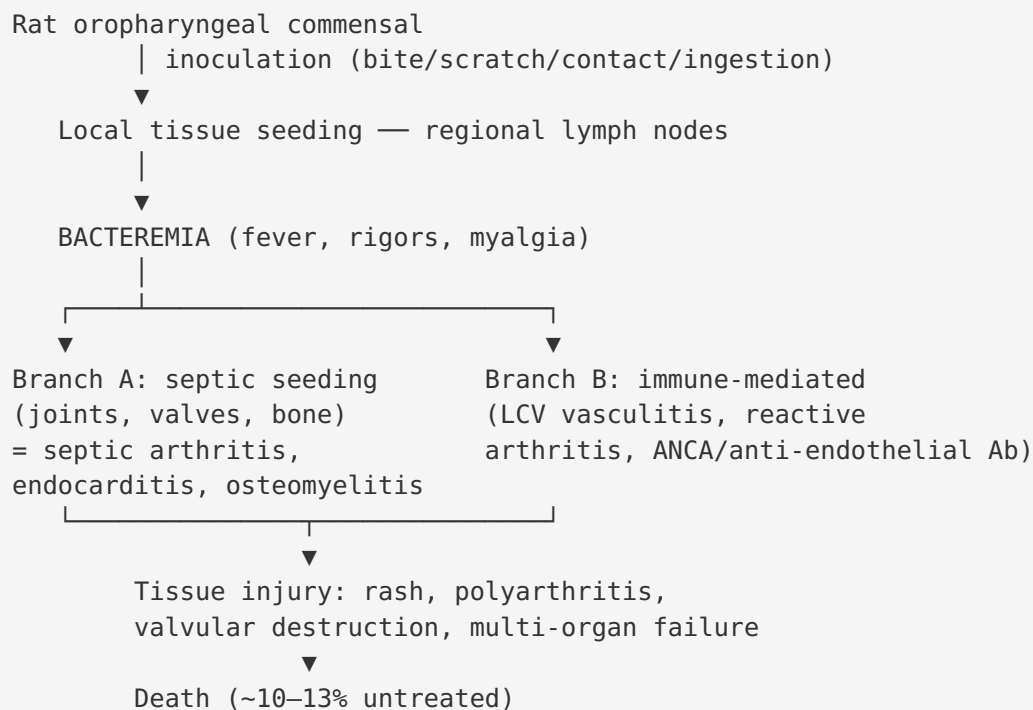
- **Infectious agents.** *Streptobacillus moniliformis* (primary) and *S. notomys* (PMID: 40472936). *Spirillum minus* causes the separate spirillary form.
 - **Environmental / occupational factors.** Contaminated rodent cages, bedding, feces, and urine; the disease can be acquired occupationally without a bite via a contaminated cage (PMID: 15029568).
 - **Transmission routes.** (1) Rodent bite or scratch; (2) direct handling / contact with excreta — "Rat-bite fever follows exposure to contaminated bodily fluids of infected rodents" (PMID: 34323361); (3) ingestion of contaminated food/water = **Haverhill fever**. Reviews confirm cases "result from rodent bites or exposure to contaminated food or water" (PMID: 42732473).
 - **Lifestyle factors.** Keeping pet rodents; occupations with rodent contact.
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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Rodent reservoir colonization** — *S. moniliformis* resides as a commensal of the rat naso-/ oropharynx → the rat becomes an asymptomatic carrier and source.
2. **Inoculation** — a bite, scratch, mucosal/skin contact with excreta, or ingestion of contaminated food/water → introduces bacteria into host tissue (or gut, for Haverhill fever).
3. **Local establishment and lymphatic spread** — bacteria seed regional tissue → drain to regional lymph nodes (sinusoidal mononuclear infiltrates observed at autopsy).
4. **Hematogenous dissemination (bacteremia)** — organisms enter the bloodstream → systemic spread and constitutional symptoms (fever, rigors, myalgia). *Documented:* autopsy shows "sinusoidal mononuclear cell infiltrates in regional lymph nodes and the liver" (PMID: 2718962).

5. **Branch A — direct septic seeding** → the bacteremia deposits organisms in joints, cardiac valves, and bone → **septic arthritis, infective endocarditis, osteomyelitis** (PMID: 40472936, PMID: 15029568).
6. **Branch B — immune-mediated injury** → bacteremia/antigenemia triggers host immune responses → **leukocytoclastic (small-vessel) vasculitis and reactive arthritis**, with induced ANCA and anti-endothelial antibodies. *Documented*: "general weakness, intermittent fever, leukocytoclastic vasculitis, and arthritis are reported," and this was "the first reported case of ANCA positivity associated with RBF" (PMID: 37450033).
7. **Convergence — tissue damage** → both branches produce mononuclear/neutrophilic infiltration and fibrinous exudate → the acral rash, migratory polyarthritis, and, in severe disease, valvular destruction and multi-organ failure.
8. **Terminal outcome (if untreated)** → progressive sepsis, endocarditis, and multi-organ failure → death in ~10–13% of untreated cases.



Cellular processes and immune involvement. Inflammation dominates, with mononuclear and neutrophilic infiltration, fibrinous endocarditis, mononuclear meningitis, hepatosplenomegaly, lymphadenopathy, and erythrophagocytosis on autopsy (PMID: 2718962). The immune-mediated branch can generate autoantibodies that mimic ANCA-associated vasculitis (PMID: 37450033).

Suggested ontology terms. GO:0006954 (inflammatory response); GO:0006935 (chemotaxis); GO:0002250 (adaptive immune response); GO:0006909 (phagocytosis, for erythrophagocytosis). Cell types: CL:0000775 (neutrophil), CL:0000235 (macrophage), CL:0000738 (leukocyte), CL:0000115 (endothelial cell, vasculitis target).

Molecular pathways / omics. No host signaling-pathway (Wnt/MAPK/mTOR/PI3K-AKT), transcriptomic, proteomic, metabolomic, or lipidomic disease signature is defined — as expected for an acute bacterial infection rather than a chronic host-driven disease. Molecular work centers on **pathogen detection** (16S rRNA, qPCR, mNGS), not host profiling.

7. Anatomical Structures Affected

Organ level (primary). Skin (rash), joints (arthritis), and blood (bacteremia). **Secondary/complication organs.** Heart valves (endocarditis; UBERON:0002134 tricuspid, UBERON:0002137 aortic, UBERON:0002135 mitral), bone (osteomyelitis; UBERON:0001474 bone tissue), meninges (UBERON:0002360), lungs (interstitial pneumonia), liver and spleen (hepatosplenomegaly), and kidneys (acute renal failure in severe sepsis).

Body systems. Integumentary, musculoskeletal, cardiovascular, hematologic/lymphatic, and — in disseminated disease — respiratory, hepatobiliary, renal, and central nervous systems.

Localization / lateralization. The rash is characteristically **acral and often bilateral** (hands and feet, including palms and soles); the arthritis is **migratory and typically asymmetric/polyarticular**.

Suggested UBERON terms. UBERON:0002097 (skin of body), UBERON:0002360 (meninges), UBERON:0001474 (bone tissue), UBERON:0000948 (heart), UBERON:0002405 (immune system), UBERON:0001456 (joint / articular structures).

8. Temporal Development

- **Incubation.** Typically **3–10 days** after exposure for the streptobacillary form (Haverhill fever similar).
- **Onset pattern.** **Acute/abrupt** — fever, chills, myalgia, followed within days by rash and migratory polyarthritis.

- **Disease course.** If untreated, a **relapsing/episodic** fever pattern can occur, or progression over days-to-weeks to endocarditis and sepsis. A fatal pet-shop case progressed to death **59 days** after the initiating injury (PMID: 15029568).
- **With treatment.** Resolution in **days to a few weeks**; the acute systemic illness is otherwise self-limited in some but unpredictably severe in others.
- **Critical period for intervention.** Early empiric antibiotic therapy (before endovascular seeding) is the decisive window; it "can prevent serious complications" (PMID: 41480582).

9. Inheritance and Population

- **Inheritance / genetics. Not applicable** — acquired infection with no heritable component, penetrance, expressivity, anticipation, founder effect, or carrier frequency.
- **Epidemiology.** Rare and underreported; not nationally reportable in most settings (PMID: 42623631). No reliable population prevalence/incidence figures exist.
- **Age distribution.** Skewed toward children — in a California reference-laboratory series, "50% of the isolates [were] from patients 9 years old or younger" (PMID: 11724672).
- **Sex ratio.** Approximately equal in the same series (roughly balanced sex distribution).
- **Vulnerable groups.** Children and people of low socioeconomic status (PMID: 34672901); occupationally exposed workers.
- **Geographic distribution.** Worldwide where rats and humans coexist; streptobacillary form predominates in the Americas and Europe, spirillary form (*Spirillum minus*) more common in Asia. The largest Canadian series reported 11 cases on Vancouver Island (2010–2016) (PMID: 31015812).

10. Diagnostics

Approach. Diagnosis requires a **high index of suspicion** anchored on the exposure history, because "Rat bite fever is a diagnosis that can be easily missed from both a clinical and a microbiological point of view" (PMID: 34672901) and "its nonspecific initial presentation combined with difficulties in culturing its causative organism produces a significant risk of delay or failure in diagnosis" (PMID: 17223620).

Microbiological detection.

Method	Performance / notes	Citation
Blood / joint-fluid culture	Gold standard but fastidious; growth may take ≥ 72 h	PMID: 38459199
SPS-aware blood culture	SPS anticoagulant inhibits growth; "up to 0.05% w/v ... seems to be inactivated, allowing for growth and detection"	PMID: 33693831
16S rRNA gene PCR/sequencing	"an even more sensitive diagnostic test" than culture; also enables species-level ID	PMID: 34672901 , PMID: 40472936
Triplex real-time qPCR	LOD 21 copies/reaction (~4–5 streptobacilli)	PMID: 35738493
Multiplex real-time PCR	1–2 genome equivalents/ μ l for <i>S. moniliformis</i> -specific target	PMID: 33618204
Metagenomic next-gen sequencing (mNGS)	"When culture is negative, 16S rRNA PCR or mNGS can be considered"	PMID: 40038603
MALDI-TOF	Can misidentify <i>S. notomys</i> as <i>S. moniliformis</i> (low score); confirm with sequencing	PMID: 40472936

Laboratory abnormalities. Nonspecific: leukocytosis/neutrophilia (or normal WBC), thrombocytopenia, elevated CRP ([PMID: 41480582](#)).

Diagnostic traps / serologic cross-reactivity. RBF has produced **false-positive dengue IgM serology** ([PMID: 41245666](#)) and **induced ANCA/anti-endothelial antibodies** mimicking ANCA-associated vasculitis ([PMID: 37450033](#)).

Differential diagnosis. Meningococcemia, disseminated gonococcal infection/gonococcal arthritis, Rocky Mountain spotted fever, secondary syphilis, infective endocarditis of other causes, reactive/viral arthritis, and systemic vasculitides ([PMID: 34323361](#)).

Genetic testing / omics diagnostics. Not applicable for host diagnosis; molecular tools target the pathogen.

11. Outcome / Prognosis

- **Untreated mortality.** RBF "carries a mortality rate of 10%" if untreated (PMID: 17223620); severe complications carry "a mortality rate as high as 13% without proper treatment" (PMID: 16858964, PMID: 42732473).
- **Complications.** Infective endocarditis (native and prosthetic valves; can require valve replacement), aortic root abscess, septic arthritis, osteomyelitis, meningitis, interstitial pneumonia, and refractory cardiogenic shock. A fatal case progressed to endocarditis "involving first the aortic valve and then the mitral valve and septum" (PMID: 15029568); another required VA-ECMO and died of refractory cardiogenic shock (PMID: 42732473).
- **Recovery with treatment.** Excellent — most patients recover fully with prompt, appropriate antibiotics.
- **Prognostic factors.** The dominant modifiable determinant is **time-to-diagnosis/treatment**; endovascular involvement (endocarditis) markedly worsens prognosis. Host genetic prognostic biomarkers do not exist.

12. Treatment

First-line pharmacotherapy. **Penicillin** is the standard therapy; RBF has an "almost omnisensitive susceptibility pattern" and responds to "a commonly available standard therapy (penicillin)" (PMID: 34672901). Penicillin G / amoxicillin are curative in uncomplicated disease.

Scenario	Regimen (reported)	Citation
Uncomplicated RBF	Penicillin G / amoxicillin	PMID: 34672901, PMID: 42623631
Severe / disseminated	Ampicillin-sulbactam + doxycycline → symptom regression	PMID: 38459199
Endocarditis / endovascular	Ceftriaxone (e.g., 6-week course)	PMID: 35693327
Atypical / localized + abscess	IV ampicillin + doxycycline + surgical debridement	PMID: 41480582

Surgical / interventional. Valve repair/replacement for endocarditis; incision-and-drainage/debridement for abscesses and septic joints; mechanical circulatory support (ECMO) in refractory shock (often a marker of late presentation).

Advanced/experimental therapeutics, pharmacogenomics, immunotherapy, gene/cell/RNA therapy. Not applicable — this is a bacterial infection managed with standard antibiotics.

Suggested NCIT terms. NCIT:C1145 (Penicillin), NCIT:C299 (Amoxicillin), NCIT:C560 (Ceftriaxone), NCIT:C561 (Doxycycline), NCIT:C1728 (Ampicillin).

13. Prevention

- **Primary prevention (behavioral/occupational).** Avoidance of rodent exposure, personal protective equipment, and rodent control. "Keeping rats as pets cannot be recommended" ([PMID: 34672901](#)); "Education on hazards of animal contact and other preventive measures are needed in small places of business like pet shops" ([PMID: 15029568](#)).
- **Wound care / post-exposure prophylaxis.** Immediate cleansing of rodent bites/scratches and consideration of post-exposure antibiotic prophylaxis (penicillin/amoxicillin) after high-risk exposures.
- **Secondary prevention.** Early empiric therapy in exposed febrile patients prevents complications ([PMID: 41480582](#)).
- **Immunization.** No licensed human vaccine exists.
- **Public-health measures.** Rodent control, education, and improved clinical awareness; because the disease is not reportable, surveillance is limited.

14. Other Species / Natural Disease

- **Reservoir.** Rats (*Rattus norvegicus*, NCBI:txid10116) carry *S. moniliformis* as a nasopharyngeal commensal. The organism "is also a pathogen in certain laboratory and domestic animals" ([PMID: 7707673](#)).
- **Natural / veterinary disease.** Causes epizootics in mice (*Mus musculus*, NCBI:txid10090), disease in guinea pigs and other species, and recurs in barrier-maintained rodent colonies;

it is an occupational hazard for laboratory-animal personnel ([PMID: 7707673](#)). Reported in a dog phlegmon.

- **Zoonotic potential.** High — pet and feeder murid rodents are documented sources of zoonotic *Streptobacillus* transmission in a global systematic review ([PMID: 37643287](#)).
- **Taxonomy.** Causative agents: *S. moniliformis* (NCBI:txid34105), *S. notomys* (distinct species; [PMID: 40472936](#)).

15. Model Organisms

Because this is a naturally occurring zoonosis, "model organisms" are the natural rodent hosts rather than engineered genetic disease models. **Laboratory mice and rats** serve as infection/carriage models and are relevant to colony health and occupational exposure ([PMID: 7707673](#)). Guinea pigs and other laboratory species are naturally susceptible. **No knockout/knock-in/transgenic host-genetic models** are relevant, since there is no host genetic etiology. Research applications center on pathogen biology, transmission, diagnostics development (e.g., qPCR/mNGS validation using clinical and animal matrices; [PMID: 35738493](#)), and colony surveillance.

Mechanistic Model / Interpretation

The disease can be understood as a two-branch dissemination model downstream of a single initiating event (rodent-to-human inoculation of a rat commensal). The **septic branch** explains the most lethal manifestations (endocarditis, osteomyelitis, septic arthritis) via direct hematogenous seeding, while the **immune-mediated branch** explains vasculitic rash, reactive arthritis, and autoantibody phenomena. Both branches converge on inflammatory tissue injury that manifests as the classic triad and, when unchecked, multi-organ failure.

Dimension	Streptobacillary RBF
Etiology	Infectious (<i>S. moniliformis</i> ± <i>S. notomytis</i>)
Host genetics	Not applicable
Key modifiable determinant of outcome	Time-to-diagnosis/treatment
Curability	High (penicillin-susceptible)
Untreated mortality	10–13%
Diagnostic bottleneck	Fastidious culture, SPS inhibition → molecular methods
Prevention	Behavioral/occupational; no vaccine

The overarching interpretation, synthesizing all 15 findings, is that **danger in RBF is a function of recognition, not host susceptibility**. The organism is nearly omnisensitive to antibiotics, so nearly all mortality is attributable to delayed diagnosis driven by nonspecific symptoms and difficult microbiology.

Evidence Base

PMID	Title (abbrev.)	Contribution
17223620	<i>Rat bite fever and Streptobacillus moniliformis</i>	Establishes organism, triad, 10% untreated mortality, diagnostic-delay risk
16858964	<i>RBF presenting with rash and septic arthritis</i>	Upper-bound 13% untreated mortality; septic arthritis
11724672	<i>RBF: a potential emerging disease</i>	Triad + exposure; pediatric age skew (50% ≤9 y)
34323361	<i>RBF: review of 29 cases</i>	Transmission via contaminated fluids; broad differential
15029568	<i>Fatal RBF in a pet-shop employee</i>	Non-bite occupational transmission; fatal endocarditis; prevention
34672901	<i>RBF: case report review</i>	16S rRNA PCR; penicillin standard; vulnerable populations; easily missed
40472936	<i>S. notomys arthritis/osteomyelitis</i>	Taxonomic expansion; sequencing > MALDI-TOF
2718962	<i>Fatal infection in a 2-month-old infant</i>	Autopsy multi-organ pathology; hematogenous dissemination
37450033	<i>RBF mimicking ANCA-associated vasculitis</i>	Immune-mediated branch; autoantibody diagnostic trap
33693831	<i>BDFX40 / 0.05% SPS 55-yr study</i>	SPS inhibition and its mitigation
35738493	<i>Triplex real-time qPCR</i>	Sensitive molecular quantification (LOD ~4–5 bacteria)
33618204	<i>Multiplex real-time PCR</i>	Species-differentiating culture-free assay
40038603	<i>Atypical RBF knee infection (China)</i>	mNGS/16S when culture negative
35693327	<i>RBF in an HIV patient</i>	Ceftriaxone for endovascular disease
42732473	<i>Fatal RBF / MCS conundrum</i>	13% mortality; endocarditis; late presentation lethal
42623631	<i>A confirmed case of RBF</i>	Underreporting; not nationally reportable
7707673		Reservoir; veterinary/laboratory-animal disease

PMID	Title (abbrev.)	Contribution
	<i>S. moniliformis</i> — a zoonotic pathogen	
37643287	Zoonotic pathogens in pet/feeder rodents	Pet/feeder rodents as zoonotic source
31015812	RBF on Vancouver Island 2010–2016	Largest Canadian series; epidemiology
41245666	False-positive dengue IgM in RBF	Serologic cross-reactivity trap; migratory polyarthritis
41480582	RBF as localized cellulitis (Nepal)	Atypical presentation; early empiric therapy prevents complications
38459199	New awareness for zoonoses / RBF	Blood culture at 72 h; ampicillin-sulbactam + doxycycline

Limitations and Knowledge Gaps

- **Evidence quality.** The literature is almost entirely **case reports and small case series**; there are no randomized trials or large prospective cohorts. Reported mortality (10–13%) derives from historical/aggregate reviews, not modern population data.
- **Underreporting.** RBF is not nationally reportable in most jurisdictions, so true incidence/prevalence are unknown and likely underestimated.
- **Pathophysiology.** The immune-mediated branch (vasculitis, ANCA induction) is documented in isolated cases; the mechanistic link (molecular mimicry vs. bystander activation) is **inferred, not experimentally demonstrated**.
- **Species attribution.** The clinical significance and relative frequency of *S. notomys* vs *S. moniliformis* are only beginning to be resolved as sequencing supplants MALDI-TOF.
- **No host-omics.** As a non-genetic infection, there are no host transcriptomic/proteomic/metabolomic signatures, biomarkers, or genetic model organisms of the disease per se.
- **Diagnostics.** Molecular assays (qPCR, mNGS) are validated largely on mock/animal matrices; broad clinical validation and availability remain limited.

Proposed Follow-up Experiments / Actions

1. **Prospective surveillance / registry.** Establish a multicenter RBF registry (given non-reportable status) to quantify incidence, exposure patterns, species distribution (*S. moniliformis* vs *S. notomytis*), and outcomes.
 2. **Molecular diagnostic deployment.** Validate and disseminate 16S rRNA PCR, triplex/multiplex qPCR ([PMID: 35738493](#), [PMID: 33618204](#)) and mNGS in routine clinical labs, and standardize SPS-aware blood-culture protocols ([PMID: 33693831](#)).
 3. **Clinical decision support.** Build an exposure-triggered alert ("febrile + rodent contact") to shorten time-to-empiric-therapy, the key outcome determinant.
 4. **Mechanistic studies.** Investigate the immune-mediated branch — whether *S. moniliformis* antigens drive ANCA/anti-endothelial antibodies via molecular mimicry ([PMID: 37450033](#)).
 5. **Occupational/public-health intervention.** Evaluate education and PPE programs for pet-shop and laboratory-animal workers, and formalize post-exposure prophylaxis guidance after rodent bites.
 6. **Species epidemiology.** Systematically re-identify archived isolates by sequencing to define the burden of *S. notomytis* human disease.
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Report compiled from 15 confirmed findings across 28 reviewed papers. Evidence source types: predominantly human clinical case reports/series and narrative reviews, with supporting in vitro/analytical diagnostic-development studies. This is an infectious, non-genetic disease; all host-genetic template sections are marked Not Applicable.
